

Functional Annotation Report: PP_0922 (AcpH) — Acyl Carrier Protein Phosphodiesterase of *Pseudomonas putida* KT2440

UniProt: Q88PC8 | **Gene / Ordered Locus:** PP_0922 | **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / NCIMB 11950 / KT2440), taxid 160488 **EC:** 3.1.4.14 | **KEGG Orthology:** K08682 | **Pfam:** PF04336 (ACP_PD) | **InterPro:** IPR007431 | **eggNOG:** COG3124

1. Summary (Answer to the Research Question)

PP_0922 encodes AcpH, an **acyl carrier protein (ACP) phosphodiesterase** (also historically called "ACP hydrolyase"; EC 3.1.4.14). Its primary catalytic function is to **hydrolyze the phosphodiester bond that links the 4'-phosphopantetheine (4'-PP) prosthetic group to the active-site serine of acyl carrier protein**, thereby converting functional **holo-ACP (or acyl-ACP) back into inactive apo-ACP** and releasing 4'-phosphopantetheine. This is the reverse of the reaction carried out by holo-ACP synthase (AcpS). AcpH is a **soluble, cytoplasmic, Mn²⁺-dependent metalloenzyme** and a non-canonical member of the HD phosphohydrolase superfamily. Biologically, it operates in the **prosthetic-group turnover / coenzyme A (CoA)-pantetheine recycling cycle** rather than in a biosynthetic step; it is dispensable for growth but required for in vivo turnover of the ACP prosthetic group. The characterized *Pseudomonas* enzyme prefers ACPs of the fatty-acid-synthesis (FAS) pathway but can also process acylated and polyketide-synthase (PKS) carrier proteins.

The gene symbol is **not ambiguous**: domain (PF04336/ACP_PD), orthology (COG3124, KO K08682), GO term (GO:0008770), KEGG EC assignment, and a full-length 39% sequence identity to the biochemically characterized *E. coli* AcpH all converge on the same function.

2. Identity Verification

Multiple independent lines of evidence confirm PP_0922 is a genuine AcpH:

Evidence type	Finding
Protein family	Single ACP_PD domain (Pfam PF04336 ; InterPro IPR007431) spanning the 194-aa protein
Orthology	COG3124 ; OrthoDB 8442777at2; KEGG KO K08682
Enzyme classification	EC 3.1.4.14 , "[acyl-carrier-protein] phosphodiesterase" (KEGG, GO:0008770)
Sequence homology	39.3% identity (75/191 aligned positions) to experimentally characterized <i>E. coli</i> AcpH (P21515), colinear over the full length with no large indels
Active-site motifs	Conserved N-terminal LAHLHL block and internal LDV..DH metallo-motif that carry the essential metal-binding residues

The organism is correctly *P. putida* KT2440 (KEGG ppu:PP_0922; gene coordinates 1,064,359–1,064,943, + strand). A proteome-wide scan shows **Q88PC8/PP_0922 is the single ACP_PD (PF04336) protein in the KT2440 genome** — AcpH is **single-copy**, so this gene is solely responsible for ACP-phosphodiesterase activity. The conserved His/Asp metal-binding constellation (His6, His8 in the LAHLHL motif; Asp78/Asp82/His83 in the LDV..FFDH motif, plus Asp49/Asp53/His55/His88/Asp94) reproduces the HD-superfamily active-site pattern whose ligands are essential in *E. coli* AcpH.

Caveat on nomenclature: UniProt's "RecName: ACP phosphodiesterase" carries a low-confidence automatic annotation source (ProtNLM/Google) and KEGG's GenBank name is "conserved protein of unknown function." However, the curated KO/EC assignments, domain architecture, orthology, and sequence conservation with a biochemically defined enzyme make the AcpH assignment robust.

3. Primary Function — The Catalyzed Reaction

Reaction (EC 3.1.4.14):



AcpH is described as "the only enzyme known to remove the 4'-phosphopantetheinyl moiety from holo acyl carrier proteins (ACP)" [PMID 20064615]. It cleaves the **phosphodiester link between 4'-phosphopantetheine and the serine-36 side chain of ACP** and is "the only enzyme known to cleave a phosphodiester-linked post-translational protein modification" [PMID 17198382]. The 4'-PP prosthetic group is the moiety onto which acyl/malonyl intermediates are loaded during fatty-acid synthesis; by removing it, AcpH inactivates ACP (apo-ACP is functionally inactive) and liberates phosphopantetheine.

Substrate specificity: - The purified *E. coli* enzyme cleaves **acyl-ACP species with acyl chains of 6–16 carbons** and is active on some, but not all, non-native ACPs [PMID 16107329] — i.e., it accepts both unacylated holo-ACP and medium/long-chain acyl-ACP. - The closely related, well-behaved ortholog *P. aeruginosa* AcpH (**PaAcpH**) — the best functional proxy for PP_0922 — shows **preferred specificity toward FAS-pathway ACPs** among eight carrier proteins tested, and it also **hydrolyzes/releases 4'-phosphopantetheinyl-linked products from a multidomain polyketide synthase**, demonstrating activity on acylated and PKS carrier-protein substrates [PMID 20064615]. - AcpH readily removes even a **modified (pantothenamide, N5-Pan) prosthetic group**, indicating tolerance for altered pantetheine moieties [PMID 20047918].

By inference for PP_0922: the physiological substrate in *P. putida* is most likely the primary FAS acyl carrier protein (AcpP, holo/acyl form), with capacity to edit/regenerate other carrier proteins.

4. Catalytic Mechanism, Structure & Fold

- **Metal dependence:** AcpH **requires Mn²⁺ for activity** [PMID 17198382].
- **Fold / superfamily:** Bioinformatic modeling placed AcpH on the structure of the **Spot hydrolase (HD) domain**, identifying it as a **non-canonical member of the HD phosphatase/phosphodiesterase superfamily** — non-canonical because it **lacks the histidine characteristic of the canonical HD motif** [PMID 17198382].
- **Active-site validation:** Putative metal-binding ligands were tested by **site-directed mutagenesis**; **removal of any of them caused severe or total loss of phosphodiesterase activity** [PMID 17198382]. In this report's alignment, the corresponding His/Asp-bearing

motifs (**LAHLHL**, **LDV..DH**) are conserved in PP_0922, supporting a shared Mn^{2+} -dependent active site.

- **Solubility:** AcpH is a soluble protein that "very readily aggregates upon overexpression in vivo or concentration in vitro" [PMID 16107329]; the *Pseudomonas* orthologs are notably **more soluble and stable** than *E. coli* AcpH, which is why PaAcpH became the tractable model [PMID 20064615].

5. Localization

AcpH is a **soluble, cytoplasmic enzyme** — it was "purified to homogeneity ... and is a soluble protein" [PMID 16107329]. It has no signal peptide or transmembrane segment (single cytoplasmic ACP_PD domain). It therefore acts in the **cytosol**, the same compartment where ACP, AcpS, the FAS-II enzymes, and CoA/pantetheine metabolism operate.

6. Biological Process & Pathway Context

The 4'-phosphopantetheine prosthetic-group cycle. ACP is functional only when its conserved serine bears a 4'-PP arm, installed from CoA by **holo-ACP synthase (AcpS / a phosphopantetheinyl transferase)** [PMID 35337986; [P 11032795](#)].

AcpH catalyzes the **opposing reaction**, removing 4'-PP to regenerate apo-ACP. Together AcpS and AcpH constitute a **prosthetic-group attachment/removal cycle**:

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apo-ACP --(AcpS, from CoA)--> holo-ACP / acyl-ACP --(AcpH, +H2O)--> apo-ACP + 4'-phosphopantetheine
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Link to CoA metabolism. Classic pulse-chase work showed the ACP 4'-PP prosthetic group **turns over metabolically**, at a rate that **depends on intracellular CoA concentration**, leading to the conclusion that "the acyl carrier protein prosthetic group turnover cycle may be related to coenzyme A metabolism rather than to lipid biosynthesis" [PMID 6363417]. Consistently, KEGG maps PP_0922 to "**Pantothenate and CoA biosynthesis**" (**ppu00770**)

within *metabolism of cofactors and vitamins*. Thus AcpH's role is best understood as **cofactor recycling / homeostasis** — returning phosphopantetheine to the CoA/pantetheine pool — and quality-control "editing" of carrier proteins, rather than a step in fatty-acid biosynthesis per se.

Non-essentiality. A total *acpH* deletion in *E. coli* grows normally, showing the activity is **not essential for growth**, "although it is required for turnover of the ACP prosthetic group in vivo" [PMID 16107329]. Apo-ACP does not normally accumulate in cells (rapidly reconverted to holo-ACP), and no apo-ACP intermediate is detected under normal or CoA-depleted conditions [PMID 6317688], underscoring that AcpH sets the balance of a dynamic cycle rather than performing an obligatory metabolic transformation.

Genomic context in KT2440. PP_0922 is transcribed **divergently** from its neighbors (PP_0920 xenobiotic reductase, PP_0921 ArsR-family regulator, PP_0923/PP_0924 ornithine-lipid acyltransferases), which all lie on the opposite strand. It is therefore **not embedded in a FAS or CoA operon**, consistent with independent regulation and a homeostatic role. (Interestingly, the adjacent ornithine-lipid acyltransferases are themselves acyl-ACP-consuming enzymes, but the opposite-strand arrangement indicates coincidental proximity rather than co-transcription.)

7. Supported and Refuted Hypotheses

Supported: 1. PP_0922 is a genuine AcpH / ACP phosphodiesterase (EC 3.1.4.14) — supported by domain, orthology, EC/KO, GO, and 39% identity to characterized *E. coli* AcpH. 2. It catalyzes hydrolytic removal of 4'-phosphopantetheine from holo-/acyl-ACP, producing apo-ACP — supported by transitive biochemical evidence from *E. coli* and *P. aeruginosa* orthologs. 3. It is a soluble, cytoplasmic, Mn²⁺-dependent HD-superfamily metallohydrolase — supported by biochemistry, modeling, and conserved catalytic motifs. 4. Its physiological role is prosthetic-group turnover linked to CoA/pantetheine metabolism, non-essential for growth.

Refuted / down-weighted: - Not an essential fatty-acid-biosynthesis enzyme (deletion is viable). - Not membrane-associated or secreted (no signal/TM; purified as soluble). - The gene is **not** part of a fatty-acid or CoA operon in KT2440.

8. Limitations & Future Directions

- **Direct biochemical data are lacking for the *P. putida* protein itself**; functional claims are transitive inferences from *E. coli* (P21515) and *P. aeruginosa* AcpH, justified by strong sequence/active-site conservation.
 - **No experimental 3-D structure** of AcpH is available (fold assignment rests on modeling onto SpoT/HD).
 - The exact in vivo substrate spectrum and any regulatory/physiological trigger for AcpH in *P. putida* (e.g., during solvent stress, PHA/rhamnolipid or secondary-metabolite production, or CoA imbalance) remain to be tested.
 - **Future work**: recombinant expression and kinetic characterization of PP_0922; substrate profiling against *P. putida* AcpP and secondary-metabolite carrier proteins; a KT2440 ΔPP_0922 phenotype under metabolic stress; and structural determination to confirm the Mn^{2+} HD-like active site.
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9. Key References

- Thomas J, Cronan JE. *The enigmatic acyl carrier protein phosphodiesterase of Escherichia coli: genetic and enzymological characterization.*

P 16107329

(2005) — identity of acpH (yajB), non-essentiality, solubility, 6–16C acyl-ACP substrate range.

- Thomas J, Rigden DJ, Cronan JE. *AcpH ... is a non-canonical member of the HD phosphatase/phosphodiesterase family.*

P 17198382

(2007) — Ser-36 phosphodiester cleavage, Mn^{2+} dependence, HD superfamily, mutagenesis of metal ligands.

- Murugan et al. *Expression, purification and characterization of the acyl carrier protein phosphodiesterase from Pseudomonas aeruginosa.*

P 20064615

(2010) — soluble Pseudomonas ortholog; FAS-ACP preference; activity on acyl/PKS carriers.

- Thomas J, Cronan JE. *Antibacterial activity of N-pentylpantothenamide...*

[P 20047918](#)

(2010) — AcpH removes modified (N5-Pan) prosthetic groups.

- Jackowski S, Rock CO. *Turnover of the 4'-phosphopantetheine prosthetic group of acyl carrier protein.*

[P 6363417](#)

(1984) — prosthetic-group turnover tied to CoA levels.

- Jackowski S, Rock CO. *Ratio of active to inactive forms of acyl carrier protein in E. coli.*

[P 6317688](#)

(1983) — apo-ACP not normally detected; rapid reconversion to holo-ACP.

- Supporting AcpS/PPTase context:

[P 35337986](#)

[P 11032795](#)

Report generated through iterative literature review (PubMed) and bioinformatic analysis (UniProt sequence retrieval, pairwise alignment to characterized orthologs, KEGG orthology/pathway and genomic-context inspection).
